2. **Map data into a consistent schema.** Raw attributes are converted into standard fields—`employmentStatus`, `deviceTrustLevel`, `onCall`, `accountTier`—ensuring that the same policy expression yields the same result across systems.
3. Assemble entity data for policy evaluation. The normalization layer returns normalized entities and attributes (principal record, resource record, relationships, and derived/contextual signals) via an API. The PDP then constructs the Cedar request—principal, action, resource identifiers, plus the context—and passes it to the evaluator.

The result isn't a single monolithic service but a shared context layer that uses a library and API used across ACME's platforms. Each PDP can request the entities it needs, confident that the underlying data will follow the same structure and freshness rules as every other system.

Some signals, like device posture, travel with the request as signed tokens issued by the device itself. Others—such as HR status or account limits—flow through the enterprise PIP. In some cases, the PIP allows ACME to simplify tokens to an identifier and scopes, with richer attributes resolved at evaluation time. Figure 12.3 illustrates how these sources come together in ACME's architecture.